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PRAVARA RURAL EDUCATION SOCIETY'S
**PRAVARA RURAL
ENGINEERING COLLEGE**
LONI
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TITLE PAGE

Name of Department : IT Subject : DBMS

Experiment No. 7

Roll No. 36

Name of Expt. Relational, Boolean, Arithmetic, Set
Operators, Group and complex Queries.

Date of Performance of Experiment :- 23/3/23

Date of Completion of Experiment :- 12/4/23

Progressive Assessment Marks

Marks Obtained				Dated Sign Teacher
Attendance (3 Months)	Performance (4 Marks)	Writing & Submission (3 Marks)	Total (10 Marks)	
3	4	3	10	



* Practical No. 7 *

Title : Relational , Boolean , Arithmetic , Set Operators , Group and Complex Queries

Objectives : Perform following SQL Queries on the databases :

- Implementation of relational operators in SQL
- Boolean operators and pattern matching
- Arithmetic operations and built-in function
- Group Functions
- Processing Data and Time Functions
- Complex queries and set operators

Theory :

1) SQL Arithmetic Operators -

- + Addition
- Subtraction
- * Multiplication
- / Division
- % Modulus



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2) SQL Comparison Operators :

=, !=, <, >, <>, >=, <=, !<, !>.

3) SQL Relational Operators :

1) AND Operator -

AND operator in SQL returns only those records which satisfy both the conditions in the SQL query.

2) OR Operator -

OR operator in SQL returns only those records that satisfy any of the condition in the SQL.

4) Logical Operators :

1) ALL -

True if all of a set of comparisons are true.

2) ANY -

True if any one of the comparison is true.



3) BETWEEN -

True if the operand is within a range.

4) EXISTS -

True if a subquery contains any rows.

5) IN -

True if the operand is equal to one of a list of expression.

6) NOT -

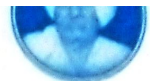
Reverse the value of any other boolean operator.

7) IS NULL -

It is used to test if the test for empty values.

8) UNIQUE -

The elimination of rows having duplicate values for specified columns.



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5) Set Operators :

- 1) UNION
- 2) UNION ALL
- 3) INTERSECT
- 4) MINUS

1) UNION -

It will be used to combine the result of two select statements.

2) UNION ALL -

This operator combines all the records from both the queries.

3) INTERSECT -

It is used to combine two SELECT statement, but it only returns the records which are common from both SELECT Statements.

4) MINUS -

It display the rows which are present in the first query but absent in the second query with no duplicates.



6) SQL DATE & TIME :

- 1) DATE - Format YYYY-MM-DD
- 2) DATETIME - Format YYYY-MM-DD HH:MI:SS
- 3) TIMESTAMP - Format YYYY-MM-DD HH:MI:SS
- 4) YEAR - Format YYYY or YY

Example :

```
SELECT DATE ("2017-06-15 09:34:21");
```

7) SQL Group Function :

These functions are -

- 1) Count()
- 2) Max()
- 3) Min()
- 4) Avg()
- 5) Sum()
- 6) Distinct()

1) Count()-

This functions returns the number of rows in the table that satisfies the



condition specified in the WHERE condition. If the WHERE condition is not specified, then the query returns the total number of rows in the table.

2) Max()-

Get the maximum value from a column.

3) Min()-

Get the minimum value from a column.

4) Avg()-

Get the average value of a numeric column.

5) Sum()-

Get the sum of a numeric column.

6) Distinct()-

Used to select the distinct rows.



Conclusion :

Thus, We have successfully executed all the operator's queries and their related functions.

①

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 9

Server version: 8.0.32 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> select * from employee_table;
```

ERROR 1046 (3D000): No database selected

```
mysql> show databases;
```

Database
bank
bank_magnt_system
bank_magt
bank_management
bank_mgnt
cust
information_schema
it
mysql
performance_schema
student
students
sys

13 rows in set (0.09 sec)

```
mysql> use bank_magnt_system;
```

ERROR 1049 (42000): Unknown database 'bank_magnt_system'

```
mysql> use bank_magnt_system;
```

Database changed

```
mysql> select * from employee_table;
```

id	name	occupation	age	address	Salary
36	naziya	student	19	shrirampur	10000
37	naz	student	18	loni	2000
38	nazi	student	18	Mumbai	10000
39	zi	student	17	delhi	3000
40	ziya	student	16	solapur	5000

5 rows in set (0.02 sec)

```
mysql> select id,salary,name,age from employee_table where salary>5000 AND age>17;
```

id	salary	name	age
36	10000	naziya	19
38	10000	nazi	18

+-----+
2 rows in set (0.01 sec)

mysql> select id,salary,name,age from employee_table where salary>10000 AND age>16;
Empty set (0.00 sec)

mysql> select id,salary,name,age from employee_table where salary<10000 AND age>16;
+-----+
| id | salary | name | age |
+-----+
| 37 | 2000 | naz | 18 |
| 39 | 3000 | zi | 17 |
+-----+
2 rows in set (0.00 sec)

mysql> select *from employee_table where salary LIKE'100%';
+-----+
| id | name | occupation | age | address | Salary |
+-----+
| 36 | naziya | student | 19 | shrirampur | 10000 |
| 38 | nazi | student | 18 | Mumbai | 10000 |
+-----+
2 rows in set (0.00 sec)

mysql> select id,salary,name,age from employee_table where salary<10000 OR age>20;
+-----+
| id | salary | name | age |
+-----+
37	2000	naz	18
39	3000	zi	17
40	5000	ziya	16
+-----+
3 rows in set (0.00 sec)

mysql> select *from employee_table where salary LIKE'200%';
+-----+
| id | name | occupation | age | address | Salary |
+-----+
| 37 | naz | student | 18 | loni | 2000 |
+-----+
1 row in set (0.00 sec)

mysql> select *from employee_table where salary LIKE'%00%';
+-----+
| id | name | occupation | age | address | Salary |
+-----+
36	naziya	student	19	shrirampur	10000
37	naz	student	18	loni	2000
38	nazi	student	18	Mumbai	10000
39	zi	student	17	delhi	3000
40	ziya	student	16	solapur	5000
+-----+
5 rows in set (0.00 sec)

mysql> select *from employee_table where salary LIKE'1_%_%';
+-----+
| id | name | occupation | age | address | Salary |


```

+-----+
| 36 | naziya | student | 19 | shrirampur | 10000 |
| 38 | nazi   | student | 18 | Mumbai    | 10000 |
+-----+
2 rows in set (0.00 sec)

```

```

mysql> select *from employee_table where salary LIKE '1__0';
+-----+
| id | name   | occupation | age | address   | Salary |
+-----+
| 36 | naziya | student    | 19 | shrirampur | 10000 |
| 38 | nazi   | student    | 18 | Mumbai    | 10000 |
+-----+
2 rows in set (0.00 sec)

```

```

mysql> select *from employee_table where salary LIKE '3_%_%';
+-----+
| id | Name | occupation | age | salary | address |
+-----+
| 39 | zi   | student    | 17 | 3000   | delhi   |
+-----+
1 row in set (0.06 sec)

```

//Nested Query

```

mysql> select *from employee_table where id IN(SELECT id from employee_table where
salary>5000);
+-----+
| id | name   | occupation | age | address   | Salary |
+-----+
| 36 | naziya | student    | 19 | shrirampur | 10000 |
| 38 | nazi   | student    | 18 | Mumbai    | 10000 |
+-----+
2 rows in set (0.01 sec)

```

```

mysql> select id,sum(salary) as total_salaries from employee_table group by id;
+-----+
| id | total_salaries |
+-----+
| 36 | 15000 |
| 37 | 2000 |
| 38 | 100000 |
| 39 | 3000 |
+-----+
4 rows in set (0.16 sec)

```

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